

In vitro to *in vivo* extrapolation modeling to facilitate the integration of transcriptomics data into genotoxicity assessment

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ABSTRACT

In vitro transcriptomics holds promise for high-throughput, human-relevant data but is not yet integrated into regulatory decision-making due to the lack of standardized approaches. For genotoxicity assessment, transcriptomic biomarkers such as GENOMARK and TGx-DDI facilitate qualitative and quantitative analysis of complex *in vitro* transcriptomic datasets. However, advancing their use in quantitative testing requires standardized methods for deriving transcriptomic Points of Departure (tPoDs) and linking them to *in vivo* responses. Herein, we investigated different approaches to calculate tPoDs and applied *in vitro* to *in vivo* extrapolation to obtain administered equivalent doses (AEDs). Human HepaRG cells were exposed for 72 h to 10 known *in vivo* genotoxicants (glycidol, methyl methanesulfonate, nitrosodimethylamine, 4-nitroquinoline-N-oxide, aflatoxin B1, colchicine, cyclophosphamide, mitomycin C, ethyl methanesulfonate, and N-Nitroso-N-ethylurea) from the highest concentration that induces up to 50 % cytotoxicity through a range of lower concentrations. Gene expression data was generated using a customized version of the TempO-Seq® human S1500 + gene panel. The GENOMARK and TGx-DDI biomarkers produced genotoxic calls for all of these reference genotoxicants. Next, we performed benchmark concentration (BMC) modeling to generate both genotoxicity-specific biomarker (tPoD-biomarkers) and generic tPoDs (tPoD_{S1500+}). High-throughput toxicokinetic models estimated the human AEDs for these tPoDs, which were compared with (a) previously reported genotoxicity-specific AEDs from other New Approach Methodologies, and (b) *in vivo* PoDs from animal studies. We found that the generic AEDs were more conservative than genotoxicity-specific biomarker AEDs. For six of the nine genotoxicants, transcriptomic AEDs were lower than the *in vivo* PoDs; refined kinetic models may improve predictions. Overall, *in vitro* transcriptomic data in HepaRG cells provide protective estimates of *in vivo* genotoxic concentrations, consistent with other *in vitro* genotoxicity testing systems.

1. Introduction

Genotoxicity testing is an important early step in chemical safety assessment. Although the existing *in vitro* regulatory genotoxicity assays are highly sensitive for hazard identification, the low specificity, low throughput, and limited mechanistic information provided by these assays have necessitated the development of new approach

methodologies (NAMs) (Ates et al., 2014; Kirkland, 2011; Menz et al., 2023; White et al., 2020). NAMs aim to reduce unnecessary animal experiments, facilitate collection of sufficient concentration-response data, increase mechanistic information needed to understand the toxicological mode of action of the compound of interest, are generally more human-relevant and higher-throughput (Kirkland et al., 2007; Menz et al., 2023; Stucki et al., 2022; Thomas et al., 2019). The

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genotoxicity-specific biomarkers GENOMARK and TGx-DDI, which consist of a subset of genes, are examples of transcriptomics NAMs that have been developed in human-derived cell lines (Ates et al., 2018; Li et al., 2017; Li et al., 2019; Thienpont et al., 2024; Thienpont et al., 2023). GENOMARK was developed in human HepaRG cells, which are becoming more often used for *in vitro* testing as this p53-competent, human-relevant cell culture model expresses relevant levels of phase 1 and phase 2 metabolic enzymes, transporters and nuclear receptors that closely resemble those of primary human hepatocytes (Seo et al., 2019). Since many genotoxic chemicals require metabolic activation to interact with and damage DNA, an *in vitro* test system that expresses the required metabolic enzymes, such as HepaRG cells, is thus needed for an accurate assessment of a chemical's genotoxicity.

Previously, we demonstrated that high-throughput transcriptomics technologies such as TempO-Seq® can be applied to identify genotoxic agents using the GENOMARK and TGx-DDI biomarkers (Thienpont et al., 2024). Different TempO-Seq® panels are available including the human S1500 TempO-Seq® panel. This widely used panel was developed by the US National Toxicology Program to enable high throughput concentration-response modeling of transcriptomic biological-response data in a cost-effective manner (Mav et al., 2018). However, the S1500 TempO-Seq® panel does not include most of the GENOMARK biomarker genes and, therefore, modified gene panels or whole genome panels should be used when running the biomarkers in parallel. The high throughput character of the TempO-Seq® technology allows the analysis of multiple concentrations, and thus supports to extend the biomarker application from hazard classification to a quantitative potency ranking using approaches such as benchmark concentration (BMC) modeling (Thienpont et al., 2024). Quantitative analysis of the *in vitro* concentration-response data with the BMC approach enables the derivation of gene-specific BMC values that can be further summarized into chemical-specific transcriptomic points of departure (tPoDs). tPoDs represent the concentration of a chemical where robust changes in the transcriptome occur. These tPoDs can either be determined based on small ('biomarker') gene sets or large ('whole transcriptome') gene sets.

The use of transcriptomic profiling for chemical screening has been proposed as the first tier of next-generation risk assessment strategies intended to replace apical endpoint-based regulatory approaches. This proposal is based on the hypothesis that concerted changes in global gene expression indicate chemical toxicity (Harrill et al., 2021; Johnson et al., 2022). Several approaches have been applied to determine a tPoD (Farmahin et al., 2017; Harrill et al., 2024; Reardon et al., 2023), but there is currently no consensus on the most appropriate one. In addition, in order to use these *in vitro* tPoDs to support the transition towards animal-free genotoxicity assessment, the relationship between *in vitro* concentration-response data and the *in vivo* response in animals/humans must be further defined.

Indeed, while *in vivo* genotoxicity doses can be compared to human exposure levels to derive a margin of exposure (Chepelev et al., 2023), this is not as straightforward for *in vitro* genotoxicity concentrations without first accounting for differences in kinetics. *In vitro* to *in vivo* extrapolation (IVIVE) tools are required to translate concentrations known to induce genotoxicity in *in vitro* assays into a dose that would elicit this response in the human body. Within this context, toxicokinetics (TK) must be considered.

High-throughput toxicokinetic (HTTK) models allow the application of IVIVE to a broad range of chemicals. These HTTK models are simplified versions of the physiologically based pharmacokinetic (PBPK) models originally developed for data-rich pharmaceuticals. By simplifying the model parameters, HTTK models can be applied to a broader range of chemicals including environmental and industrial chemicals for which, in most cases, less information is available (Cohen Hubal et al., 2019; Wambaugh et al., 2018; Wetmore, 2015). The HTTK models generally use TK data (e.g., intrinsic hepatic clearance (Cl_{int}) and plasma protein binding) from *in vitro* assays, as well as structure-derived physicochemical properties and species-specific physiological data to predict

a steady-state concentration (C_{ss}) *in vivo* (Pearce et al., 2017). By taking into account TK information and the concept of reverse dosimetry (Wetmore, 2015), HTTK models can be used to translate the genotoxic chemical-specific PoD measured *in vitro* into a dose expected to elicit genotoxicity *in vivo*, providing a human dose context for *in vitro* results (Beal et al., 2023; Friedman et al., 2020). The administered equivalent dose (AED) (in mg/kg bw/day) generated by these HTTK models thus represents the estimated dose that needs to be administered *in vivo* to reach a steady-state concentration in the plasma that is equal to the concentration inducing genotoxicity in the *in vitro* system.

For genotoxicity, three case studies have shown the utility of IVIVE using HTTK modeling for deriving AEDs from *in vitro* genotoxicity data (Beal et al., 2023; Beal et al., 2024; Kuo et al., 2022). The first case study was limited to *in vitro* micronucleus (MN) data (Kuo et al., 2022), whereas the second case study included different types of novel *in vitro* genotoxicity tests (further referred to as NAMs) and compared the *in vitro* AEDs to PoDs from relevant animal studies (Beal et al., 2023). Both studies concluded that for most chemicals, *in vitro* AEDs are more protective for human health than PoDs derived from animal data. The final study (Beal et al., 2024), used the yH2AX/pH3 assay to evaluate DNA damage *in vitro* of 13 trichothecenes and applied IVIVE to derive AEDs that were compared to human exposure data to assess risk potential.

The overall goal of our study was to investigate different approaches that can be used to derive protective *in vitro* tPoDs for application in risk assessment and regulatory decision-making, providing data for future development of best practices. Specifically, we explored which mathematical approach and types of gene-expression dataset (i.e. hazard-specific versus hazard-agnostic) lead to protective tPoDs (i.e., tPoDs at or below apical PoDs). We also investigated potential regulatory applications of these tPoDs through the use of HTTK modeling to derive transcriptomic AEDs. To do this, we produced genotoxicity-specific tPoDs from a subset of genes (i.e., GENOMARK and TGx-DDI biomarkers) as well as 'generic' tPoDs derived from a larger set of transcriptomic data ('customized S1500 + panel'). We investigated two mathematical approaches to derive genotoxicity-specific biomarker tPoDs: the bootstrap median BMC and the 5th percentile BMC (Bourdon et al., 2013; Rowan-Carroll et al., 2021; Thienpont et al., 2024). To derive generic tPoDs, we used three different approaches: the 25th lowest ranked gene BMC, the lowest consistent response dose (LCRD), and the lowest pathway BMC based on REACTOME pathways (Reardon et al., 2023). HTTK modeling was applied to derive transcriptomic AEDs and to compare our genotoxicity-specific biomarker AEDs to generic AEDs based on a larger (customized S1500 + panel) gene set. Finally, to support scientific confidence in our results, the transcriptomic AEDs were also compared to genotoxicity-specific AEDs derived from other *in vitro* tests and *in vivo*-based PoDs as published previously (Beal et al., 2023). Overall, our work aims to increase confidence in the use of potency metrics such as tPoDs and AEDs from *in vitro* transcriptomic data in regulatory risk assessment, particularly in instances when apical or *in vivo* data are not available.

2. Materials and methods

2.1. Chemicals

Gene expression data for ten chemicals that are established to be genotoxic (GTX) *in vivo* were used in this study. In addition, to confirm that the HepaRG cells are effectively expressing the required metabolic enzymes, three chemicals that require metabolic activation were included as this is important in assay validation using HepaRG cells for genotoxic effects (Table 1):

- (i) Eight GTX chemicals for which TempO-Seq® data had been collected previously (Thienpont et al., 2024): glycidol (GLY), methyl methanesulfonate (MMS), nitrosodimethylamine (NDMA), 4-nitroquinoline-N-oxide (4NQO), aflatoxin B1 (AFB1),

Table 1
Overview of the test chemicals^a.

Chemical and CAS number	<i>In vitro</i> GTX		<i>In vivo</i> GTX	Requires metabolic bioactivation	Mechanism of GTX	Concentrations tested (μM)					TempO-Seq panel used ^b	TempO-Seq data extracted from a previous study?
	Ames	MN/CA				C1	C2	C3	C4	C5		
aflatoxin B1 (AFB1) - (1162–65–8)	+	+	+	Yes	formation of DNA adducts and ROS	0.025	0.1	0.5	1.5	NA	(1)	Yes (Thienpont et al., 2024)
colchicine (COL) - (64–86–8)	-	+	+	No	aneugen	0.1	0.3	1	3	NA	(1)	Yes (Thienpont et al., 2024)
cyclophosphamide (CPA) - (6055–19–2)	+	+	+	Yes	alkylation	78	233	700	2100	NA	(1)	Yes (Thienpont et al., 2024)
ethyl methanesulfonate (EMS) - 62–50–0	+	+	+	No	alkylation	411	1233	3700	4500	5000 *	(2)	No, newly generated
glycidol (GLY) - (556–52–5)	+	+	+	No	alkylation	97	291	873	1200	1402 *	(2)	Yes (Thienpont et al., 2024)
methyl methanesulfonate (MMS) - (66–27–3)	+	+	+	No	mono functional alkylation	27	81	243.3	400	500 *	(2)	Yes (Thienpont et al., 2024)
mitomycin C (MMC) - (50–07–7)	+	+	+	No	mono and bifunctional alkylation	0.17	0.5	1.5	4.5	NA	(1)	Yes (Thienpont et al., 2024)
nitrosodimethylamine (NDMA) - (62–75–9)	+	+	+	Yes	alkylation and DNA adduct formation	80	240	720	2160	5454 **	(2)	Yes (Thienpont et al., 2024)
N-nitroso-N-ethylurea (ENU) - 759–73–9	+	+	+	No	alkylation	183	550	1650	2125	2600 *	(2)	No, newly generated
4-nitroquinoline-N-oxide (4NQO) - (56–57–5)	+	+	+	No	formation of DNA adducts and ROS	0.9	2.7	8	12 *	NA	(1)	Yes (Thienpont et al., 2024)

^a Chemicals are shown with names and abbreviations, CAS number, previously published *in vitro* and *in vivo* genotoxicity data, mechanisms of action, and the concentrations tested (μM). GTX denotes genotoxicity, defined as positive in the Ames test, the micronucleus assay (MN) or via analysis of chromosomal aberrations (CA). NA = not available, only four concentrations tested. *Maximum concentration due to cytotoxicity (IC₅₀). **The highest stock solution possible was prepared at a concentration of 1090 mM in DMSO. ROS = reactive oxygen species

^b Legend: (1) the TempO-Seq® analysis was conducted using a customized version of the human S1500 + panel using four concentrations of the chemical, whereas for (2) the analysis was done using the human whole transcriptome panel using five concentrations of the chemical.

colchicine (COL), cyclophosphamide (CPA), and mitomycin C (MMC). Test chemicals were purchased from Merck/Sigma-Aldrich (Hoeilaart, Belgium).

- (ii) Two additional GTX chemicals for which TempO-Seq® data were newly generated: ethyl methanesulfonate (EMS) and N-Nitroso-N-ethylurea (ENU). The two chemicals were purchased from Merck/Sigma-Aldrich (Hoeilaart, Belgium). ENU was dissolved in dimethylsulfoxide (DMSO) and further diluted in the medium with a final concentration of 0.5 % (vol/vol) DMSO, whereas EMS was directly dissolved in medium.

For the TempO-Seq® experiments, concentrations were based on the results of the cell viability assays. The concentration inducing a maximum of 10 % cytotoxicity (i.e., IC₁₀) value was selected as a guide (C3, the third concentration within the concentration range) to derive the concentration range to be tested. Lower and higher concentrations around the IC₁₀ value were then included, resulting in concentration intervals of 3-fold (an exception is noted for AFB1, for which the concentration range was based on previous qPCR experiments). Cases where the concentrations above C3 deviated from the 3-fold concentration distribution were 4NQO, EMS, GLY, MMS, and ENU. For these chemicals, the highest concentration (C4 in case of 4NQO and C5 for EMS, GLY, MMS and ENU) induced 50 % cytotoxicity and, therefore higher concentrations could not be included. In all cases, the top concentrations induced a maximum of 50 % cytotoxicity (i.e., IC₅₀) as per the established protocol used for TGx-DDI (Buick et al., 2020, 2021), or, if there was no cytotoxicity, a top concentration of 10 mM or the limit of solubility was used.

2.2. Collection of TempO-Seq® data

TempO-Seq® sequencing data reported in Thienpont et al. (2024) were used for the eight previously tested genotoxins. New data were generated for the two additional genotoxins (ENU and EMS) using an identical protocol to Thienpont et al. (2024). Every experiment was performed in triplicate using different batches of HepaRG cells,² which were purchased from Biopredic International. HepaRG cells were used for both (i) cell viability measurement using a CellTiter-Glo Luminescence Viability Assay kit³ (cat. no. G7571; Promega Corporation, Madison, USA) and (ii) generation of cell lysates for the TempO-Seq® analysis (n = 3 per treatment group) as described in (Thienpont et al., 2024). The TempO-Seq® analysis was conducted by BioClavis (Glasgow, UK), a spinout company of BioSpyder Technologies. The TempO-Seq® Human Whole Transcriptome Reagent Kit (v2.1) (BioSpyder Technologies, Carlsbad, CA, USA) was used (comprising 22,537 probes over 19,687 genes) to prepare libraries in a 96-well plate format from exposed and control HepaRG cell lysates, according to the manufacturer's instructions and as previously described (Yeakley et al., 2017). As discussed in the previous study (Thienpont et al., 2024), two different TempO-Seq® panels were used to collect the transcriptomic data (i.e., the customized S1500 + panel, which refers to the TempO-Seq®

² <https://www.heparg.com/rubrique-differentiated-heparg-cells-hpr116>

³ Technical bulletin 288: https://be.promega.com/-/media/files/resources/p/rotocols/technical-bulletins/0/celltiter-glo-luminescent-cell-viability-assay-protocol.pdf?rev=30e8ec640fdd4866b207e28c0bc497c&sc_lang=en

S1500 + panel (Mav et al., 2018) comprising 3456 probes targeting 2792 genes customized to include all GENOMARK genes, and the human whole transcriptome panel (Matteo et al., 2023; Ramaiahgari et al., 2019)). Therefore, the output of the TempO-Seq® Human Whole Transcriptome assays were scaled to the customized S1500 + panel to ensure a comparable analysis of the chemicals (cfr. Section 2.4). It should be noted that previously the robustness of the S1500 + platform was evaluated and demonstrated that this panel is an acceptable surrogate for the whole transcriptome (Lee et al., 2021; Mav et al., 2018).

2.3. Gene expression biomarker analysis: GENOMARK and TGx-DDI classification

Statistical modeling and bioinformatics tools as previously described (Thienpont et al., 2024) were used to classify EMS and ENU as (i) GTX or NGTX using the GENOMARK biomarker or (ii) DDI (DNA damage inducing) or non-DDI (NDDI) using the TGx-DDI genomic biomarker. In brief, for both the GENOMARK and TGx-DDI biomarker genes, counts per million-normalized read counts were $\log_2$ transformed. In the case of GENOMARK (84 genes), two classifiers based on supervised machine learning algorithms were applied: (1) the support vector machine (SVM) using R packages e1071 and caret; and (2) the random forest (RF) algorithm using the randomForest package in R Cran, Version 4.0.4 as described previously (Thienpont et al., 2023). The output of both the SVM and RF classifier is a probability value between 0 and 1 for genotoxicity. A chemical is classified overall as GTX when the probability is > 0.55 in both classifiers and as non-genotoxic when the probability is < 0.45 in both classifiers. When the probability values are between 0.45 and 0.55 or when the chemical produced different results in both classifiers, the chemical is marked as inconclusive. For TGx-DDI (64 genes), three separate analyses were conducted to classify the chemicals: (1) the Nearest Shrunken Centroids (NSC) using the pamr function of R; probability analysis (PA; visualized using heatmaps); (2) principal component analysis (PCA) using the prcomp() function in R; and (3) hierarchical clustering (HC) using the hclust() function in R (www.r-project.org). Classification was completed as follows: if a chemical yielded a DDI call in any one of three classification analyses (NSC PA heatmaps, PCA, or HC), it was classified as DDI; whereas, a chemical was classified as non-DDI if it did not have any DDI calls in any of the aforementioned analyses.

2.4. Benchmark concentration modeling and transcriptomic point of departure derivation

To conduct the BMC analysis, $\log_2$ normalized read counts were uploaded into BMDExpress v3, in accordance with recommendations outlined in the US NTP Approach to Genomic Dose-Response Modeling report (NTP, 2018; Phillips et al., 2019). Modeling was performed on the GENOMARK and TGx-DDI biomarker gene sets (84 and 64 genes, respectively) or for the entire customized S1500 + panel gene set. Gene expression data were analyzed and filtered using the Williams trend test, retaining features with a permutation p-value < 0.01 (with 500 permutations) and an absolute value of linear fold change (FC) > 1.5 for at least one concentration relative to matched solvent controls. To derive the BMC confidence intervals (CIs), the genes that passed the prefilters were fit to the following models: Linear, Exponential (3 and 5), Polynomial 2, and restricted Power (power restricted to ≥ 1). To identify the best-fitting model in BMDExpress v3, a nested chi-square test is first applied to select the optimal polynomial degree for the dose-response relationship. Following this, Akaike Information Criterion (AIC) is used to confirm the best model among those selected, balancing goodness-of-fit with model complexity.

The benchmark response was set to 1 SD based on the recommendations from the expert panel held by the US NTP (NTP, 2018). BMCs were filtered based on a goodness-of-fit criterion, retaining only those with a p-value > 0.1 , a BMC/BMCL ratio < 20 , a BMCU/BMCL ratio

< 40 , and the BMC $<$ the highest concentration (where BMCL and BMCU represent the lower and upper confidence limits of the BMC, respectively).

After BMC modeling, two types of tPoD values were derived: (a) genotoxicity-specific tPoDs using the smaller GENOMARK and TGx-DDI biomarker gene sets, or (b) generic tPoDs using the larger customized S1500 + panel gene set. Currently, there is no consensus yet on best practices for tPoD derivation. Therefore, we used approaches that were applied in previous studies and that we reference below.

a. Collection and calculation of the genotoxicity-specific biomarker tPoD values

For the calculation of the genotoxicity-specific biomarker tPoDs, two approaches were used:

1. The bootstrap median BMC

This approach is based on the inclusion of all the modeled gene BMCs derived from the BMC modeling. For 8 out of the 10 chemicals, the biomarker bootstrap median BMC values (tPoD_{GENOMARK,med} and tPoD_{TGx-DDI,med}) were collected from (Thienpont et al., 2024).

For the 2 additional test chemicals (i.e., EMS and ENU), tPoD values for GENOMARK and TGx-DDI were derived by calculating the bootstrap median BMC using the same protocol as described in (Thienpont et al., 2024). From the resulting 2000 simulated experiments, the distribution of the median BMC was estimated and the 95th percentile CIs for the bootstrap median BMCs (noted as tPoD_{TGx-DDI,med} and tPoD_{GENOMARK,med}) were defined. GraphPad Prism 9.3.0. (GraphPad Software Inc.) was used to $\log_{10}$ transform the BMCL and BMCU values and to visualize the CIs of each chemical in graphs.

2. The 5th percentile ranked BMC

Since we assumed that the bootstrap median approach would be less conservative than data-distribution approaches for the derivation of tPoDs, a second approach (the 5th percentile ranked BMC) was applied to generate more conservative tPoDs_{GENOMARK,5th} and tPoDs_{TGx-DDI,5th}. We calculated the BMC at the 5th percentile of all biomarker genes affected by substance treatment as a conservative value for tPoD derivation (tPoD_{GENOMARK,5th} and tPoD_{TGx-DDI,5th}) (Bourdon et al., 2013; Friedman et al., 2020; Reardon et al., 2023; Rowan-Carroll et al., 2021). Unlike the median approach, which only considers genes that fit BMCs, the 5th percentile approach takes into account the gene set size. For TGx-DDI, the 5th percentile represents the 3rd ranked BMC in the gene set as TGx-DDI has 64 genes. For GENOMARK, the 5th percentile BMC is the 4th ranked BMC as there are 81 genes in the gene set. Similar to the bootstrap median BMC, 95 % confidence intervals were generated using the bootstrap percentile approach. Thus, bootstrap distributions were obtained for the 3rd ranked BMC for TGx-DDI and the 4th ranked BMC for GENOMARK and the 2.5th percentile (BMCL) and the 97.5th percentile (BMCU) from these distributions were used to estimate the 95 % CI of the BMC.

b. Calculation of generic tPoD values

For the 10 genotoxic chemicals in this study, generic tPoDs were derived by applying BMC modeling on the human customized S1500 + panel. The $\log_2$ normalized read counts per chemical were reduced to include only the genes from the customized S1500 + panel to ensure a comparable analysis of the chemicals.

After BMC modeling, three approaches previously studied in Reardon et al. (2023) were used to derive generic tPoDs (a comparison of the three approaches can be found in Table 2):

1. 25th lowest ranked gene BMC

This approach is based on data distribution (i.e., the ranking of genes). All genes with BMCs were ranked from lowest to highest BMC value. The BMC of the 25th ranked gene was selected as the

Table 2

Summary of advantages and disadvantages of the three approaches (25th lowest-ranked gene, lowest consistent response dose (LCRD), and the REACTOME approach) used for tPoD derivation based on the customized S1500 + panel.

Approach	Advantages	Disadvantages
25th lowest-ranked gene	- Simplified calculation using the 25th ranked gene by potency based on the modeled BMCs. - No additional software or complex modeling is needed after the BMC analysis. - Eliminates chemicals that show insufficient biological activity (i.e., lacking 25 concentration-responsive genes)	- Not widely used yet as the selection of the 25th gene is considered arbitrary. - Still requires establishing standards within the expert community to define a cut-off gene.
LCRD	- Typically the lowest and most conservative tPoD of all approaches. - Based on the distribution of BMCs across all chemicals.	- Not widely used yet. - May still capture tPoDs in the noise range or below the lowest concentration.
Lowest median pathway BMC using the REACTOME Pathways database	- The most used tPoD. - Endorsed by the NTP as a reliable method for deriving tPoD values that reflect meaningful changes in biology while reducing the influence of background (NTP, 2018). - Shown to correlate with apical endpoints in <i>in vivo</i> studies (Gwinn et al., 2020; Thomas et al., 2013).	- tPoDs may vary depending on the pathway database used. - Randomized gene sets produce tPoDs similar to those from curated pathway databases, however little biological information is used in the randomized gene set approach (Costa et al., 2024). - May result in wider confidence intervals compared to other approaches (Addicks et al., 2023).

Adapted from ((Costa et al., 2024; Reardon et al., 2023).

concentration at which the test chemical induced a concerted change in the transcriptome (Johnson et al., 2022; Matteo et al., 2023; Reardon et al., 2023). An advantage of this approach is that chemicals that are inactive (i.e., do not have at least 25 responsive genes) are eliminated.

2. The lowest consistent response dose (LCRD)

The LCRD approach is a data-driven approach (Crizer et al., 2021) that identifies the lowest dose level where a consistent response in biological features is observed. The BMC values were first ranked from lowest (rank 1) to highest. For each BMC value, a ratio was calculated (BMC of $n + 1$ / BMC of rank n). This ratio represents the relative degree of change in the BMC between each ranked feature. The LCRD was identified as the lowest BMC from which subsequent BMCs had ratio values less than $\frac{1}{4}$ log difference in values (<1.66). Thus, this reflects the concentration at which transcriptional changes are occurring within similar concentrations. Also, low BMCs (ratios > 1.66) that may be biological outliers are eliminated (Crizer et al., 2021). It is hypothesized that the exposure levels at or above the LCRD are more likely to be toxicologically relevant due to the more consistent rate of change as a function of dose in the biological system.

3. The lowest pathway BMC based on REACTOME Pathways

The REACTOME-based BMC approach uses pathways to filter out noise in the data and ensure that the tPoD is based on a biologically relevant gene set (NTP, 2018). This approach was defined by the United

States National Toxicology Program (US NTP). To capture the most sensitive pathway (i.e., the pathway with the lowest BMC), genes and their corresponding BMC values were matched to their associated pathways using the REACTOME Pathways database (Fabregat et al., 2017). Pathways where at least 5 % of the genes had BMCs, and representing at least three genes, were included. The lowest BMC median was selected from these pathways. It should be noted that the function of the selected pathway is not considered when interpreting the data.

For each of these three approaches, the bootstrap distribution was used to estimate the median BMC and to generate CIs. GraphPad Prism 9.3.0. (GraphPad Software Inc.) was used to $\log_{10}$ transform the BMCL and BMCU values and to visualize the CIs of each chemical in graphs. The bootstrap median BMCs_{S1500+} were used as tPoDs_{S1500+} for HTTK modeling.

2.5. High-throughput toxicokinetic (HTTK) modeling to convert tPoD values into administered equivalent doses (AED)

IVIVE modeling to estimate the AEDs in mg/kg bw/day was performed using the httk R package (version 2.3.0) (Pearce et al., 2017). A three-compartment steady-state model in R (i.e., the ‘3compartmentss’ model) was used to estimate the concentration at steady state (C_{ss}) in plasma for a dose rate of 1 mg/kg bw/day. The relationship between C_{ss} and dose rate is assumed to be linear, and thus, at a modeled dose rate of 1 mg/kg bw/day the calculation of AED simplifies to AED = tPoD/C_{ss} using Equation 1.

Equation 1: Calculation of the Administered Equivalent Dose (AED) in mg/kg bw/day.

$$AED(\text{mg/kg bw/day}) = tPoD(\mu\text{M}) \times \frac{1\text{mg/kg/day}}{C_{ss}(\mu\text{M})}$$

AED values were calculated for both the genotoxicity-specific biomarker tPoDs (tPoD_{GENOMARK_med}, tPoD_{TGX-DDI_med}, tPoD_{GENOMARK_5th} and tPoD_{TGX-DDI_5th}) and the three generic tPoDs_{S1500+} (i.e., tPoD_{25th_gene}, tPoD_{LCRD}, and tPoD_{REACTOME}). A similar approach to the one described by Beal et al. (2023) was used for the IVIVE modeling. More specifically, to run the ‘3compartmentss’ model, the R function `calc_mc_css` was employed using default parameters with the exception of `output.units = "uM"`. The httk package provides a Monte Carlo (mc) sampling tool that represents the uncertainty and inter-individual variability within the sampled population (Ring et al., 2017). Therefore, the R function ‘`calc_mc_css`’ models the upper 95th percentile steady-state concentration, i.e., the C_{ss,95}. The C_{ss,95} accounts for individuals with higher plasma concentrations from the same exposure in order to obtain a scaling factor for correlating *in vitro* concentrations with AED. The ‘3compartmentss’ model requires the input of different *in vitro* TK parameters and physicochemical properties: fraction unbound in the plasma protein (F_{unbound}), intrinsic clearance (Cl_{int}), octanol/water partition coefficient (logP), and molecular weight (MW). Afterwards, the modeled C_{ss} was used to estimate the AED (mg/kg bw/day). For two of the case study chemicals, there was sufficient data for the required input parameters within the httk library to model C_{ss} values. For the remaining chemicals with any missing input TK parameters, *in silico* predictions were obtained using the ADMET Predictor (version 10) tool, following previously used methods (Beal et al., 2022).

2.6. Published NAM AED and *in vivo* PoD data extraction

Genotoxicity-specific AED values based on other *in vitro* tests and *in vivo* PoDs previously reported (Beal et al., 2023) were compared to the transcriptomic AEDs estimated in this study. Beal et al. (2023) included 31 chemicals with at least one source of *in vitro* genotoxicity data and *in vivo* animal data. The *in vitro* genotoxicity tests included the *in vitro* transgenic rodent (TGR) gene mutation assay, ToxTracker® (Bcl2 and Rtnk reporter genes only), Prediscreen (γH2AX and phospho-histone H3), Multiflow® (γH2AX, p53, p-H3, and polyploidy), and the *in vitro*

Microflow® flow cytometry version of the MN assay (OECD, 2023). *In vivo* assays included the MN assay (OECD, 2016), fluorescence *in situ* hybridization assay, chromosomal aberration assay (OECD, 2014), aneuploidy assays, TGR assay (OECD, 2022), Pig-a-assay, hypoxanthine phosphoribosyl transferase (HPRT) assay and Thymidine Kinase assay. They performed BMC and BMD modeling in the PROAST software using a benchmark response (BMR) of 100 on these data set (Boisvert, 2020). When *in vivo* dose-response data could not be modeled, the lowest-observed-genotoxic-effect-levels (LOGELs), indicative of significant genotoxicity responses that were reported by the identified studies, were used as *in vivo* PoDs. AEDs for *in vitro* tests were calculated by dividing the compound-specific BMC₁₀₀ in the respective *in vitro* method by the estimated C_{ss} predicted with the htkk R package (version 2.2.1) (Pearce et al., 2017).

A log₁₀ difference was calculated based on Equation 2 below. This was adapted from Friedman et al. (2020) and Reardon et al. (2023) and used to compare the median AEDs from *in vitro* transcriptomic data to median AEDs from other *in vitro* tests and *in vivo* PoDs extracted from (Beal et al., 2023). For this analysis, a median AED_{biomarkers} was calculated based on the four genotoxicity-specific biomarker AED values and one median AED_{S1500+} was calculated based on the 3 generic AEDs derived for the larger gene set. This log₁₀ difference has been used to determine if *in vitro* tPoDs are conservative estimates of *in vivo* effects for numerous substances across a large chemical space (Reardon et al., 2023). A log₁₀ difference < 0 indicates that the AED is greater than the PoD_{in vivo}, suggesting that the *in vitro* assay may not be sufficiently protective, while a log₁₀ difference > 0 expressed that the AED was less than the PoD_{in vivo} (Friedman et al., 2020). A log₁₀ difference within 1 order of magnitude on the log₁₀ scale, which corresponds to a 10-fold difference (i.e., 10¹) on a linear scale, is typically assumed as an acceptable value for *in vitro* tPoDs as conservative estimates of *in vivo* values given the range of variability observed across *in vivo* studies (Ly Pham et al., 2020). In addition, to convert the log₁₀ differences into fold differences on a linear scale, the absolute median log₁₀ difference was used.

Equation 2: Calculation of the log₁₀ difference (mg/kg bw/day) based on the difference of the median log₁₀ PoDs and median log₁₀ AEDs.

$$\text{Log}_{10}\text{difference} = \text{median log}_{10}\text{PoD}_{\text{in vivo}} - \text{median log}_{10}\text{AED}_{\text{in vitro transcriptomics}}$$

Table 3
Predicted classification as genotoxic (GTX), inconclusive (INCONCL), or non-genotoxic (NGTX) by GENOMARK, or as DNA damaging (DDI), non-DNA damaging (NDDI), or unclassified (INCONCL)^a by TGx-DDI^b.

Chemical	Historical in vivo genotoxicity data	Concentration (µM)	GENOMARK			TGx-DDI			
			RF	SVM	Overall Classification	PCA	HC	NSC	Overall Classification
EMS [‡]	+	411	0.38	0.27	NGTX	NDDI	NDDI	NDDI	NDDI
		1233	0.61	0.59	GTX	DDI	DDI	DDI	DDI
		3700	0.88	0.98	GTX	DDI	DDI	DDI	DDI
		4500	0.88	1.00	GTX	DDI	UNCL	DDI	DDI
		5000	0.88	0.99	GTX	DDI	UNCL	DDI	DDI
ENU [‡]	+	183	0.3	0.17	NGTX	NDDI	NDDI	NDDI	NDDI
		550	0.64	0.45	INCONCL	DDI	DDI	DDI	DDI
		1650	0.92	0.99	GTX	DDI	UNCL	DDI	DDI
		2125	0.87	1.00	GTX	DDI	UNCL	DDI	DDI
		2600	0.89	0.99	GTX	DDI	UNCL	DDI	DDI

^aUNCL = unclassified. Unclassified samples had profiles that did not cluster with either DDI or NDDI reference compounds. ^bAbbreviations for classification methods: random forest (RF), support vector machine (SVM), principle component analysis (PCA), hierarchical clustering (HC), nearest shrunken centroids (NSC). [‡]EMS = ethyl methanesulfonate, ENU = N-Nitroso-N-ethylurea

3. Results

3.1. Hazard classifications

The biomarker classifications of the chemicals published previously (Thienpont et al., 2024) are in supplementary Table A.1.

Prior to transcriptome analyses, cell viability studies were performed for EMS and ENU to select a top concentration for the Tempo-Seq® experiments that induced a maximum of 50 % cytotoxicity (Supplementary Fig. A.1). The GENOMARK and TGx-DDI prediction models were used to classify EMS and ENU as GTX or NGTX, or as DDI or non-DDI, respectively (Table 3). Both *in vivo* GTX chemicals were accurately classified as GTX using GENOMARK and as DDI using TGx-DDI. Both biomarkers identified the lowest concentration of EMS and ENU as non-GTX. For GENOMARK, a concentration-dependent increase was seen in the probability for genotoxicity with both the random forest (RF) and the support vector machine (SVM) models. For ENU, an inconclusive result was found at 550 µM using the SVM model and thus, at 550 µM, the overall call given is inconclusive using GENOMARK. However, this concentration is classified as DDI using TGx-DDI. These discrepancies may be explained by the algorithm behind the SVM model. The concentration of 550 µM likely induces a weak genotoxic effect under the investigated conditions. Indeed, the RF model classifies ENU at a concentration of 550 µM as GTX, suggesting it may be marginally more sensitive to weak signals in GENOMARK analyses. Nevertheless, to classify the genotoxic hazard of ENU overall, we take all the concentrations into account. Thus, since ENU is classified as GTX at higher concentrations, in both GENOMARK and TGx-DDI, it was classified as GTX.

3.2. Comparison of the potency rankings using biomarker tPoDs versus generic tPoDs

As EMS and ENU were classified as GTX and DDI by GENOMARK and TGx-DDI, the corresponding tPoD_{GENOMARK,med}, tPoD_{GENOMARK,5th}, tPoD_{TGx-DDI,med} and tPoD_{TGx-DDI,5th} values and their associated derived confidence intervals. An overview of these values as well as the ones for the eight genotoxic chemicals (i.e., GLY, MMS, NDMA, 4NQO, AFB1, COL, CPA, MMC) previously reported in (Thienpont et al., 2024) is

provided in [supplementary Table A.2.1](#) and [Table A.2.2](#) containing the bootstrap median derived tPoDs and 5th percentile derived tPoDs, respectively. Within this previous study, we showed that similar potency rankings were obtained using the bootstrap median tPoDs for either GENOMARK or TGx-DDI (potency rankings are based on the lowest bootstrap median representing the most potent chemical to the highest, least potent chemical). When repeating this analysis with an extended dataset, including data from EMS and ENU, this finding was again confirmed. In addition, a comparison of the potency ranking between the bootstrap median tPoDs versus the 5th percentile ranked tPoDs resulted led to two observations. First, a similar potency ranking was obtained for the 10 genotoxics ([Fig. 1](#)). Second, the 5th percentile ranked BMC approach produced lower (i.e., more conservative) tPoDs than the bootstrap median approach, except for EMS ([Fig. 1](#)). However, no firm conclusion can be made on potency differences across the approaches since their CIs overlap ([Wills et al., 2016](#)). We note that the broad width of the CIs for COL relative to the other GTX compounds was discussed in detail in our previous paper ([Thienpont et al., 2024](#)).

BMC modeling on the customized S1500 + panel allowed the derivation of generic tPoDs. For each chemical, an overview of the total number of genes fitting the BMC models and the estimated bootstrap median BMC (i.e., tPoD) with its corresponding BMCL and BMCU was calculated for the three different approaches (25th lowest ranked gene, LCRD and REACTOME) ([Table 4](#)). The top two most transcriptionally active chemicals by this analysis (i.e., chemicals with the most genes fitting BMC models) were ENU and EMS, with 2726 and 5870 genes fitting models, respectively. In contrast, NDMA was the least transcriptionally active with only 413 genes fitting models.

Chemicals were ranked according to their toxicological potency (lowest to highest BMCs) using three generic tPoDs: (1) the 25th lowest-ranked gene BMC (i.e. tPoD_{25th_gene}), (2) LCRD (i.e. tPoD_{LCRD}), and (3) REACTOME lowest pathway median BMC (i.e., tPoD_{REACTOME}) ([Table 4](#)). The potency rankings for the 10 chemicals based on the three generic tPoDs were highly similar ([Fig. 1](#)). More specifically, the potency ranking based on the tPoD_{25th_gene} was: COL > AFB1 > MMC > 4NQO > MMS > GLY > CPA > NDMA > ENU > EMS. However, the CIs of two chemicals (i.e., CPA and NDMA) overlapped as well as those of EMS and ENU. Consequently, no firm conclusion on the ranking between these compounds can be made. The potency ranking based on the tPoD_{LCRD} was: COL > AFB1 > MMC > 4NQO > MMS > GLY > NDMA > CPA > ENU > EMS. Overlapping CIs were observed for AFB1, COL, and MMC; for MMS and GLY; and for CPA, NDMA, EMS, and ENU. The tPoD_{REACTOME} potency ranking was: COL > AFB1 > MMC > 4NQO > MMS > GLY > CPA > NDMA > ENU > EMS. Here, the CIs overlapped for COL, AFB1, and MMC; for MMS and GLY; and for CPA, NDMA, ENU and EMS. Overall, when comparing the three approaches, the 25th lowest-ranked gene approach had the smallest CIs, with BMCU/BMCL ratios ranging from 1.35 to 1.82 ([Fig. 1](#) and [Table 4](#)). The BMCU/BMCL ratios of the LCRD approach ranged from 2.76 to 5.80, whereas for the lowest REACTOME pathway approach ranged from 2.74 to 19.61.

Importantly, the potency rankings based on the generic tPoDs were also highly similar to the ones based on the genotoxicity-specific biomarker tPoDs ([Fig. 1](#)). To compare the tPoDs between the biomarkers and the large gene set, the median tPoDs were calculated (i.e., median tPoD_{biomarkers}, calculated based on the two tPoD derivation approaches for both GENOMARK and TGx-DDI, $n = 4$; and the median

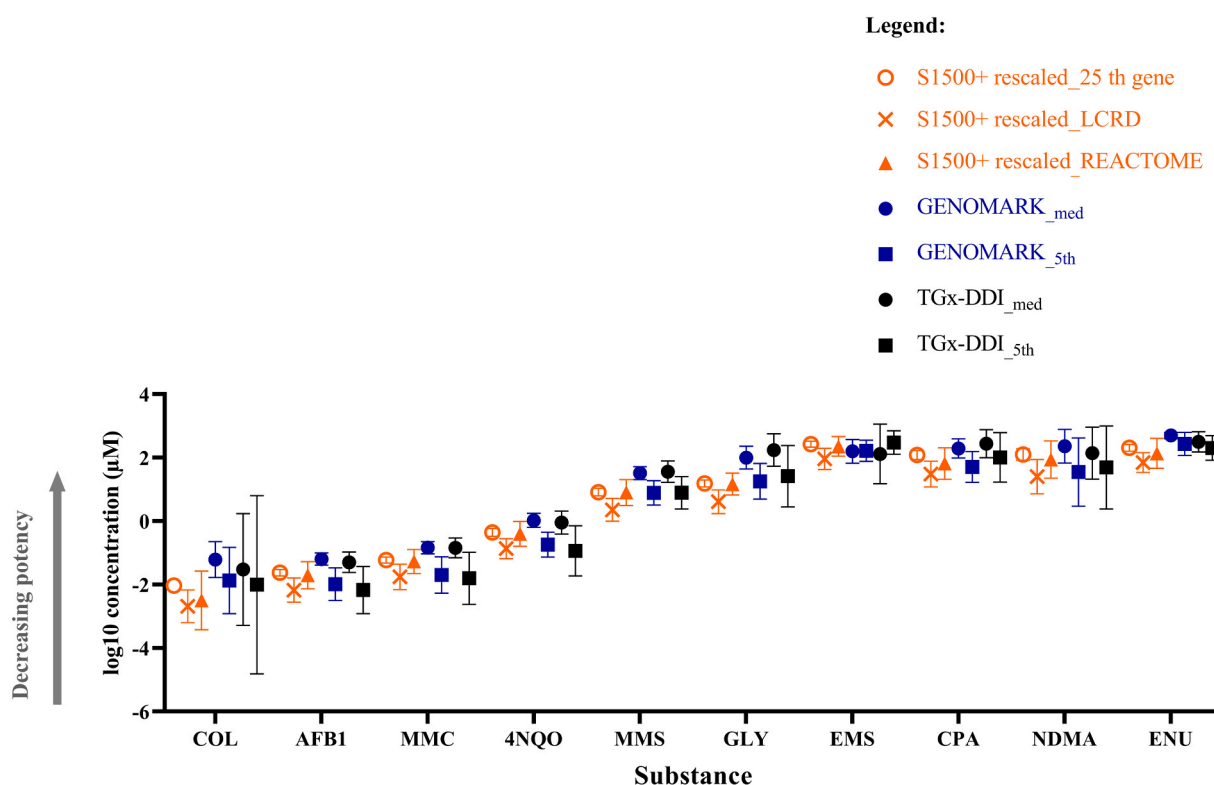


Fig. 1. Potency comparison using the GENOMARK or TGx-DDI biomarker genes and the large set of genes (rescaled S1500 + panel) with bootstrap median benchmark concentrations (BMCs). Median BMCs are shown as $\log_{10}$ concentration (μM) on the y-axis with their lower and upper 95 % confidence limits. Potency measures are shown for the TGx-DDI and GENOMARK transcriptomic biomarker (in dark grey and blue, respectively). Chemicals were ranked from left to right according to their GENOMARK biomarker bootstrap median BMC: potency rankings are from the lowest bootstrap median, representing the most potent chemical, to the highest, least potent chemical. The potency ranking from the rescaled S1500 + panel (in orange) for the 25th lowest-ranked gene (circle), lowest consistent response dose (LCRD) (cross), and REACTOME (triangle) approach are also shown. Chemicals: glycidol (GLY), methyl methanesulfonate (MMS), nitrosodimethylamine (NDMA), 4-nitroquinoline-N-oxide (4NQO), aflatoxin B1 (AFB1), colchicine (COL), cyclophosphamide (CPA), mitomycin C (MMC), ethyl methanesulfonate (EMS) and N-Nitroso-N-ethylurea (ENU).

Table 4
Comparison of transcriptomic point of departures of the large gene set (tPoDs_{S1500+}) generated by the 25th lowest ranked-gene (tPoD_{25th_gene}), lowest consistent response dose (LCRD) (tPoD_{LCRD}), or lowest REACTOME pathway approach (tPoD_{REACTOME})^a.

Chemical	Genes with at least 1 Bootstrap BMC	25th lowest ranked-gene		LCRD		REACTOME pathway	
		tPoD _{25th_gene} (BMCL - BMCU) (μM)	Ratio (BMCU/ BMCL)	tPoD _{LCRD} (BMCL - BMCU) (μM)	Ratio (BMCU/ BMCL)	tPoD _{REACTOME} (BMCL - BMCU) (μM)	Ratio (BMCU/ BMCL)
COL	1372	0.009 (0.007–0.012)	1.66	0.002 (0.001–0.005)	5.46	0.007 (0.001–0.014)	19.61
AFB1	2030	0.024 (0.02–0.028)	1.4	0.01 (0.004–0.012)	3.45	0.02 (0.01–0.04)	4.05
MMC	2097	0.059 (0.051–0.069)	1.37	0.018 (0.009–0.034)	3.73	0.061 (0.029–0.098)	3.43
4NQO	1875	0.443 (0.36–0.533)	1.48	0.142 (0.081–0.227)	2.8	0.481 (0.211–0.754)	3.57
MMS	1413	8.265 (6.717–9.967)	1.48	2.392 (1.257–4.054)	3.23	9.713 (4.050–15.495)	3.83
GLY	1332	15.366 (12.750–18.229)	1.43	4.169 (2.209–7.480)	3.39	15.554 (8.373–25.629)	3.06
CPA	1355	120.909 (95.895–151.358)	1.58	33.865 (15.57–58.650)	3.77	71.673 (29.045–145.605)	5.01
NDMA	413	125.2 (92.899–169.005)	1.82	23.029 (10.424–60.476)	5.8	101.378 (33.184–226.232)	6.82
ENU	2726	201.662 (169.331–243.274)	1.44	77.058 (42.185–116.262)	2.76	162.632 (62.959–291.073)	4.62
EMS	5870	268.078 (228.792 – 308.167)	1.35	99.293 (52.840–156.153)	2.96	251.544 (136.222–372.989)	2.74

^a Chemicals are ordered in the table based on the tPoD_{25th_gene} values. Each chemical (glycidol (GLY), methyl methanesulphonate (MMS), nitrosodimethylamine (NDMA), 4-nitroquinoline-N-oxide (4NQO), aflatoxin B1 (AFB1), colchicine (COL), cyclophosphamide (CPA) and mitomycin C (MMC), ethyl methanesulphonate (EMS), N-Nitroso-N-ethylurea (ENU)) was modeled using a BMR of 1 SD to derive BMCs. Bootstrap median BMCs (i.e., tPoDs) are given together with their lower and upper confidence intervals (BMCL and BMCU, respectively).

tPoD_{S1500+}, calculated of the three individual tPoDs_{S1500+}, n = 3). Overall, the tPoDs based on the large gene set were more conservative (i. e., lower) than the biomarker tPoDs, except for EMS (Fig. 1 and Table 5). For the remaining 9 genotoxicants, examination of the ratio of the median tPoD_{biomarkers}/ median tPoD_{S1500+} showed that the tPoDs for the large gene set were 1.35- to 5.69-fold (in case of COL) lower than the biomarker tPoDs (Table 5).

To explore where in the full distribution of gene BMCs the biomarker genes position (i.e., the sensitivity of the genes to the chemicals), we produced gene accumulation plots of the top 1000 gene BMCs for the 10

genotoxic chemicals and annotated the ranks of the TGx-DDI and GENOMARK biomarker genes (Fig. 2). The cut-off of the 25th gene is marked as a blue dashed line in the graph. This analysis revealed that several GTX-specific genes (i.e., biomarker genes) were activated at low

Table 5
Comparison of median transcriptomic point of departure (tPoD) values using a benchmark response of 1 standard deviation based on the biomarker (GENOMARK and TGx-DDI) gene sets and the customized S1500 + panel for the ten genotoxicants used in this study^a.

Chemical	Median tPoD _{biomarker} (μM)	Median tPoD _{S1500+} (μM)	Ratio median tPoD _{biomarker} / median tPoD _{S1500+}
AFB1	0.0375	0.02	1.88
COL	0.06	0.01	5.69
MMC	0.085	0.06	1.42
4NQO	0.595	0.44	1.35
MMS	21.1075	8.27	2.55
GLY	76.7675	15.37	4.99
NDMA	157.745	101.38	1.56
CPA	176.83	71.67	2.47
EMS	201.92	251.54	0.80
ENU	323.93	162.63	1.99

^a Chemicals are ordered in the table based on the median tPoD_{biomarker} values. The median tPoD_{biomarker} was derived from two different approaches for both GENOMARK and TGx-DDI: the bootstrap median BMC approach and 5th percentile-ranked BMC. The median tPoD_{S1500+} was derived from three different approaches, namely the 25th lowest ranked-gene, lowest consistent response dose (LCRD) or REACTOME approach. Chemicals included: glycidol (GLY), methyl methanesulphonate (MMS), nitrosodimethylamine (NDMA), 4-nitroquinoline-N-oxide (4NQO), aflatoxin B1 (AFB1), colchicine (COL), cyclophosphamide (CPA), mitomycin C (MMC), ethyl methanesulphonate (EMS) and N-Nitroso-N-ethylurea (ENU).

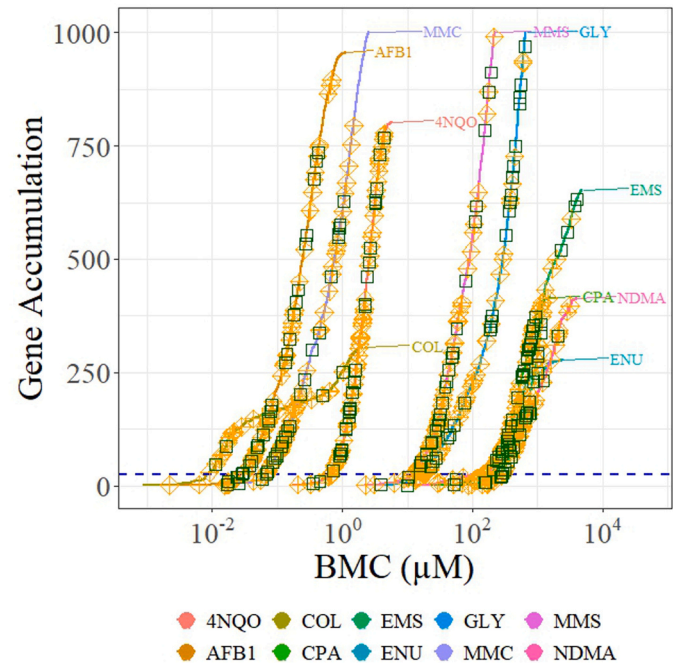


Fig. 2. Distribution of TGx-DDI and GENOMARK biomarker genes within the gene accumulation plot of the customized S1500 + panel for the ten genotoxicants. Biomarker genes are shown as orange squares for GENOMARK and as dark green squares for TGx-DDI. The x-axis indicates each gene's BMC value (μM) and the y-axis ranks the number of genes from the lowest to the highest BMC within the top 1000 genes within the customized S1500 + panel. The blue dashed line represents the cut-off of the 25th gene.

concentrations. More specifically, for the 10 genotoxicants, 2–10 genes of the 25 genes equal or below the 25th gene were GENOMARK genes (supplementary table A.3). In contrast, the number of TGx-DDI genes found among these 25 genes varied between 0 and 3 per genotoxicant (Supplementary Table A.3).

3.3. Comparison of transcriptomic AEDs to other *in vitro* AEDs and *in vivo* PoDs

The values for the different TK parameters used for IVIVE modeling of the 10 test chemicals (log P, Cl_{int} and $F_{unbound}$) can be found in supplementary Table A.4. Only 2 out of the 10 chemicals were present in the httk database: AFB1 and COL. For 8 out of the 10 chemicals, the missing TK parameters were estimated based on *in silico* predictions using the ADMET v10 software. The *in silico* prediction of Cl_{int} was outside the domain of applicability of ADMET v10 for MMS, NDMA, and GLY. Consequently, these Cl_{int} predictions are associated with a high uncertainty and should be used with caution. Table 6 provides a summary of the chemical-specific estimated steady-state concentration (Css_{95}) and AED (mg/kg bw/day) for GENOMARK, TGx-DDI, and the customized S1500 + panel.

Next, transcriptomic AEDs were compared to the previously reported *in vitro* AEDs or *in vivo* PoDs from (Beal et al., 2023). Except for glycidol, all chemicals were also included in the study and the corresponding log₁₀ AEDs and log₁₀ PoDs for these compounds were retrieved from the manuscript from Beal et al. (2023). It should be noted that the *in vitro* data from Beal et al. (2023) used BMC₁₀₀ values whereas in our study we applied a BMR of 1 SD which is a different approach to derive the BMC values. Also, many of the *in vivo* BMD₁₀₀ values were not useful for PoD derivation due to limitations in the animal study design (i.e., limited dose selection). Therefore, LOGELs were also used to derive the *in vivo* median log₁₀ PoDs. Supplementary Fig. A.2 illustrates the distribution of the individual chemical-specific *in vitro*-based AEDs (transcriptomic AEDs derived in this study and previously reported *in vitro* AEDs from Beal et al. (2023) and *in vivo* PoDs for the 9 genotoxicants.

To account for the effects of outliers, limitations in dose/concentration range, and the variety of genotoxicity endpoints measured in the assays, the median log₁₀ AED and PoD quantiles were calculated similarly to the approach by Beal et al. (2023) (Fig. 3). An overview of both the previously reported median log₁₀ *in vitro* AEDs and the median log₁₀ PoDs derived from animal assays, as well as the median log₁₀ biomarker or generic AEDs derived from the transcriptomic data, can be found in Table 7, together with the number of assays used to derive these values. To compare transcriptomic AEDs with *in vivo* PoDs, the log₁₀ difference was calculated. For the majority of chemicals (6/9 chemicals), the log₁₀ difference AED_{S1500+} was > 0, indicating that these AEDs were more conservative (i.e., lower) than animal-derived PoDs. The three

chemicals that had a median log₁₀ AED_{S1500+} equal or higher than the median log₁₀ *in vivo* PoD were CPA, ENU, and NDMA. For CPA and NDMA, the absolute median log₁₀ difference was 0 and 0.07, respectively, corresponding to a median AED_{S1500+} 1-fold (i.e., 10⁰) and 1.17-fold (i.e., 10^{0.07}), respectively, higher than the median *in vivo* PoD. For ENU, an absolute log₁₀ difference of 1.68 was observed, corresponding to a median AED_{S1500+} value 47.86-fold higher than the median *in vivo* PoD.

Similar to the log₁₀ difference based on the AED_{S1500+}, most genotoxic chemicals (6/9) had median log₁₀ AED_{biomarkers} values lower than the median log₁₀ *in vivo* PoD (Table 7 and Fig. 3). CPA and NDMA had higher median log₁₀ AED_{biomarkers} values (2.1-fold and 1.2-fold higher, respectively) than the median log₁₀ *in vivo* PoDs. However, ENU had a very high median log₁₀ difference of 1.94 from the *in vivo* PoD value, corresponding to a 87.7-fold increase over the *in vivo* PoD.

4. Discussion

In vitro transcriptomics is increasingly being applied to evaluate chemical toxicity. Today, the higher-throughput gene expression technologies (e.g., TempO-Seq®) enable the generation of large toxicogenomic datasets for application in hazard and risk assessment. Transcriptomic biomarkers facilitate toxicogenomic data analysis and provide an objective and efficient approach to identify potential hazards from complex whole transcriptomic datasets. Furthermore, BMC modeling enables the derivation of tPoDs for a variety of regulatory purposes. However, despite their potential to generate high-throughput and human-relevant data, transcriptomic NAMs have not yet been fully integrated into regulatory hazard and risk assessment (Meier et al., 2024). This is primarily due to questions surrounding the most suitable approaches to apply these omics data in regulatory decision-making. More specifically, there is no consensus yet on which tPoD derivation strategy and which tPoD should be used for quantitative risk assessment. Herein, we explored several approaches to tPoD derivation, compared hazard-specific and hazard-agnostic tPoDs, and compared the AEDs (which are obtained after extrapolation of these tPoDs) to previously published genotoxicity-specific *in vitro* and *in vivo* PoDs using IVIVE.

We first investigated how biomarker tPoDs derived from smaller gene sets (i.e., genotoxicity-specific biomarker tPoDs) compare to generic tPoDs derived from a larger gene set. For genotoxicity-specific tPoDs, we initially selected the bootstrap median BMC based on our previous study (Thienpont et al., 2024) that focused on potency ranking. The bootstrap is an excellent tool for complex statistics like BMC modeling applied on transcriptomics data. Moreover, we previously (Thienpont et al., 2024) demonstrated that the bootstrap median BMC resulted in highly similar potency rankings between GENOMARK and TGx-DDI. However, for risk assessment, it is preferable to be sufficiently

Table 6

Summary of the estimated upper 95th percentile steady-state concentration (Css_{95}) and the chemical-specific administered equivalent dose (AED) based on the tPoD values for GENOMARK, TGx-DDI and the custom S1500 + panel.

Compound	httk Css_{95} (μM)	GENOMARK _{med}	GENOMARK _{5th}	TGx-DDI _{med}	TGx-DDI _{5th}	S1500 + _{25th_gene}	S1500 + _{LCRD}	S1500 + _{REACTOME}
		AED (mg/kg bw/day)	AED (mg/kg bw/day)	AED (mg/kg bw/day)	AED (mg/kg bw/day)	AED (mg/kg bw/day)	AED (mg/kg bw/day)	AED (mg/kg bw/day)
4NQO	7.645	0.139	0.028	0.116	0.028	0.058	0.019	0.063
AFB1	11.43	0.006	0.001	0.005	< 0.001	0.002	0.001	0.002
COL	7.22	0.009	0.002	0.011	0.008	0.001	< 0.001	0.001
CPA	2.869	63.445	23.233	114.159	45.698	42.143	11.804	24.982
NDMA*	8.094	38.345	4.541	28.936	6.132	15.468	2.845	12.525
GLY*	7.634	12.554	2.502	19.355	5.812	2.013	0.546	2.037
MMS*	5.813	5.603	1.550	5.944	1.427	1.422	0.412	1.671
MMC	3.417	0.043	0.007	0.043	0.006	0.017	0.005	0.018
EMS	4.214	40.199	39.722	44.180	67.567	63.616	23.563	59.693
ENU	4.986	103.444	53.677	65.463	37.288	40.446	15.455	32.618

* Chemicals that need to be interpreted with caution as the *in silico* prediction of Cl_{int} was outside the domain of applicability of ADMET v10

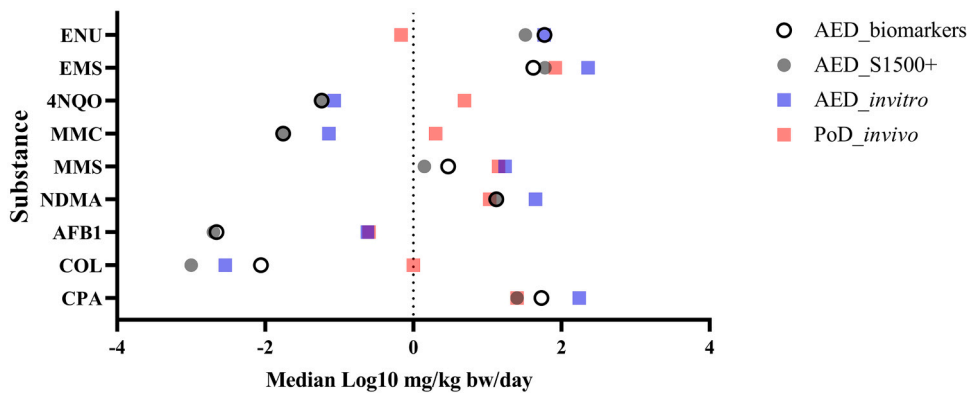


Fig. 3. Comparison of administered equivalent doses (AEDs) from (1) biomarkers (GENOMARK and TGx-DDI), (2) customized S1500 + panel, (3) previously published in vitro studies and (4) point of departures (PoDs) from in vivo studies (Beal et al., 2023) using median log₁₀ values. The median transcriptomics AEDs (open black and font grey circles) were compared against the median in vitro AEDs (blue squares) and in vivo PoDs (red squares). The dashed line represents the point where the median log₁₀ = 0.

Table 7

Overview of the median biomarker log₁₀ administered equivalent doses (AED_{biomarkers}) and the median log₁₀ AED_{S1500+} derived in this study and the publicly available in vitro median log₁₀ AED and in vivo median log₁₀ point of departures (PoDs) for genotoxicity (Beal et al., 2023)^a.

Chemical	Median log ₁₀ AED _{biomarkers}	Median log ₁₀ AED _{S1500+}	Median log ₁₀ AED _{in vitro}	Median log ₁₀ PoD	Log ₁₀ difference	
	(mg/kg bw/day)	(mg/kg bw/day)	(mg/kg bw/day); (number of NAMs included)	(mg/kg bw/day); (Number of in vivo assays included)	Median log ₁₀ PoDs versus median log ₁₀ AED _{S1500+} ; (Linear scale fold difference)	Median log ₁₀ PoDs versus median log ₁₀ AED _{biomarkers} ; (Linear scale fold difference)
CPA	1.73	1.4	2.24; (6)	1.4; (19)	0; (1)	-0.33; (2.14)
COL	-2.06	-3	-2.54; (6)	0; (17)	3; (1000)	2.06; (115.41)
AFB1	-2.66	-2.7	-0.62; (7)	-0.6; (7)	2.1; (125.89)	2.06; (114.74)
NDMA	1.12	1.1	1.65; (1)	1.03; (1)	-0.07; (1.17)	-0.09; (1.24)
MMS	0.47	0.15	1.24; (9)	1.15; (9)	1; (10)	0.68; (4.79)
MMC	-1.76	-1.76	-1.14; (10)	0.3; (19)	2.06; (114.82)	2.06; (115.63)
4NQO	-1.24	-1.24	-1.07; (9)	0.69; (20)	1.93; (85.11)	1.93; (85.18)
EMS	1.62	1.76	2.36; (6)	1.83; (24)	0.07; (1.18)	0.21; (1.60)
ENU	1.77	1.51	1.77; (16)	-0.17; (50)	-1.68; (47.86)	-1.94; (87.68)

^a The log₁₀ difference was calculated for comparison of the transcriptomics AEDs for either (1) the AED_{S1500+} based on the log₁₀ median PoD – log₁₀ median AED_{S1500+} difference or either (2) the AED_{biomarkers} based on the log₁₀ median PoD – log₁₀ median AED_{biomarkers} difference

protective. Therefore, we also applied the 5th percentile BMC as a more conservative approach compared to the bootstrap median. The selection of the 5th percentile BMC was based on other studies investigating data-distribution approaches including the 5th percentile BMC (Bourdon et al., 2013; Reardon et al., 2023; Rowan-Carroll et al., 2021) to increase the sensitivity in the calculation of the BMC. Moreover, the 5th percentile BMC is comparable to the approach used previously for ToxCast data (Friedman et al., 2020). Since there is no consensus yet on which approach to use for deriving a hazard-specific tPoD, we compared the two approaches (i.e. the bootstrap median BMC and 5th percentile BMC) to best provide a protective tPoD. Our findings confirm that the 5th percentile BMC results in more conservative tPoDs, and therefore, we proposed this approach to be more suitable for the derivation of protective hazard-based tPoDs.

We found that generic tPoDs yielded potency rankings that were very similar across the three approaches (the 25th lowest-ranked gene, LCRD, and the lowest REACTOME pathway BMC). However, in line with previous observations (Reardon et al., 2023), the LCRD consistently produced the lowest tPoDs for all 10 genotoxic chemicals, whereas the tPoDs derived with the other two approaches were very similar. The BMC CIs, were the smallest for the 25th lowest-ranked gene approach. Furthermore, the 25th lowest-ranked gene approach resulted in the least overlap of CIs between chemicals. This means that in our study, the 25th ranked gene approach was most suited to identifying potential differences in potencies across the chemicals since the potencies of chemicals are only different when their CIs do not overlap (Wills et al., 2016).

Other studies (Matteo et al., 2023; Reardon et al., 2023; Reardon et al., 2021) investigating tPoDs have proposed that the 25th gene BMC is a reliable predictor of the concentration at which the concerted molecular response in a transcriptomic data set occurs and thus provides a protective tPoD. Our data suggest that for genotoxicants, the three approaches yielded similar tPoD values for the specific (S1500 +) gene set, but the 25th lowest ranked-gene approach may be preferred given the precision of the estimate indicated by the smaller CIs.

When comparing the generic and biomarker tPoDs to explore how potency rankings might differ using hazard-agnostic versus hazard-based approaches for the 10 genotoxicants, the potency rankings were similar overall, using these two approaches. This suggests that both approaches could be used for potency ranking of chemicals. However, the ratio of median tPoDs (tPoD_{biomarkers}/tPoD_{S1500+}) showed that the generic tPoDs were at least 1.35-fold lower than the specific tPoDs, except for EMS. These results suggest that the hazard-based (i.e., biomarker) approach to derive a tPoD is less conservative than the hazard-agnostic (i.e., generic) approach. However, a direct comparison of the biomarker versus hazard-agnostic tPoDs is rather difficult since a different approach was used to derive the hazard-agnostic (gene-distribution or pathway-based) and hazard-based tPoDs (initially, a median gene BMC was derived from the biomarker gene sets (Thienpont et al., 2024)). Several genes in these biomarker gene sets are responding in the same concentration range and below the 25th ranked genes of the S1500 + panel, whereas the GENOMARK genes appear more frequently among the top 25 ranked genes. The higher number of GENOMARK

genes in the top 25 ranked genes may be because the data were collected under the experimental conditions used for the development of GENOMARK (i.e. exposure duration of 72 h in HepaRG cell line (Ates et al., 2018) compared to a 4 h exposure period in TK6 cells for TGx-DDI (Li et al., 2017). However, it is important to stress that despite this observation, the tPoDs derived based on either the GENOMARK or TGx-DDI biomarker genes were very similar. Selection of the median may not be the most conservative approach to tPoD derivation for a transcriptomic biomarker; specifically, the median of the gene set is biased to a higher tPoD compared to the hazard-agnostic approaches that select the most sensitive genes and pathways. Indeed, in this study, the 5th percentile ranked BMC approach resulted in more conservative genotoxicity-specific tPoDs that were closer to the generic tPoDs. Overall, our findings are consistent with observations in the literature. For example, (Matteo et al., 2023) found that hazard-agnostic tPoDs based on the 25th lowest-ranked gene BMC approach were more conservative than hazard-specific tPoDs derived with a transcriptomic biomarker for estrogenic activities using the median BMC value. Further work is needed to arrive at consensus on the most suitable tPoD for risk assessment. This will require a collaborative and international effort and is a research priority of different working groups amongst which are the European Partnership for the Assessment of Risks from Chemicals (PARC) consortium and the Health and Environmental Sciences Institute (HESI) tPoD working group (O'Brien et al., 2025). Practically, it is unlikely that a universal tPoD would suffice across all potential outcomes, or serve as the sole endpoint (Reardon et al., 2023). Instead, a range of tPoD derivation strategies should be considered across diverse *in vitro* datasets.

To enable a comparison of the *in vitro* transcriptomic concentration-response data to *in vivo* data, IVIVE modeling was used to estimate external doses (i.e., transcriptomic AEDs) that correspond to the *in vitro* concentration that induces a biological effect in the *in vitro* transcriptome assays. We found that the median $\log_{10}$ generic AEDs were generally lower (i.e., more conservative) than the median $\log_{10}$ biomarker AEDs. In addition, we compared the biomarker AEDs to the genotoxicity-specific *in vitro* AEDs and *in vivo* PoDs. The median $\log_{10}$ difference between *in vitro* AED_{biomarkers} and *in vivo* PoDs was > 0 for six of the nine genotoxic chemicals, except for NDMA, CPA, and ENU (i.e., $\log_{10}$ AED_{biomarkers} were lower than the *in vivo* PoDs). This observation is in line with the results of other *in vitro* genotoxicity studies (Beal et al., 2023; Kuo et al., 2022) showing that *in vitro* genotoxicity AEDs are typically conservative relative to *in vivo* outcomes. For example, Kuo et al. (2022) found that AEDs derived from *in vitro* MN data were protective of human health for the majority of genotoxic chemicals tested (24/33 chemicals). Beal et al. (2023) used a broader set of NAMs and found that AEDs were health-protective relative to *in vivo* PoDs from animal studies in 20 of the 31 chemicals they studied (Beal et al., 2023). Despite the limited number of chemicals in our study and variability in the number of *in vivo* studies used to produce apical PoDs, our results support previously studies showing that genotoxicity-specific AEDs in HepaRG cells are more conservative than *in vivo* genotoxic PoDs for the majority of chemicals. Since the generic transcriptomic AEDs were more conservative than the biomarker AEDs, these values were also lower than the *in vivo* genotoxicity-specific PoDs for the majority (6/9) of chemicals. This aligns with results from other studies comparing generic *in vitro* transcriptomic AEDs to *in vivo* PoDs. For example, Chen et al. (2022), derived AEDs for perfluorooctane sulfonate (PFOS) from human microarray *in vitro* data (Chen et al., 2022). They observed that AEDs were generally lower than PoDs from mouse *in vivo* data that evaluated multiple endpoints. Reardon et al. (2023) analyzed human TempO-Seq® data across different test chemicals, cell culture models, and exposure times for tPoD determination. They observed that the generic transcriptomic AEDs were similar or lower than the PoDs for three apical endpoints (developmental toxicology, and reproductive toxicology) for the majority of chemicals tested. These observations suggest that generic tPoDs based on the most sensitive gene expression and pathway

responses provide a protective alternative to apical endpoint PoDs from animal studies. Our results point further in the direction that hazard-agnostic transcriptomic-based AEDs are similar or more protective when compared to genotoxicity-specific NAM AEDs or *in vivo* PoDs, but more studies using additional cell lines and a larger chemical selection are needed to confirm this.

It is important to understand cases where median AEDs are not lower than the *in vivo* PoDs to ensure that AEDs are protective of human health when applied to risk assessment applications. The median $\log_{10}$ AED values based on both biomarkers or the larger transcriptome panel were higher than the median $\log_{10}$ *in vivo* PoDs for three chemicals: CPA, NDMA, and ENU. This might be due to limitations with the generic HTTK approach used, and for these chemicals, more complex models or other considerations in data interpretation may be required for IVIVE implementation (Beal et al., 2023b). In the case of CPA and NDMA, the median $\log_{10}$ AEDs were slightly higher than the median $\log_{10}$ *in vivo* PoDs. Nevertheless, the difference was less than 1 on a $\log_{10}$ scale, corresponding to a 10-fold difference on a linear scale, indicating that these AEDs are very close to the apical PoDs. Previous studies have shown that there is 10-fold variability in the PoDs produced between animal studies (Ly Pham et al., 2020). Thus, to be within 10-fold difference is consistent with expectations. The transcriptomic AED values for these two compounds are thus on the same order of magnitude as the *in vivo* based PoDs. In contrast, ENU had a median $\log_{10}$ AED – PoD difference of two orders of magnitude, which is in line with previous observations (Beal et al., 2023) for ENU. In addition, the median biomarker AED for ENU was within the range of the median AED value of other *in vitro* assays. ENU is a potent mutagen but is unstable at neutral pH (Arcos et al., 2013) and is hydrolyzed rapidly. Therefore, the stated concentration of ENU in the test medium, which is used for the AED calculation, may be higher than the real concentration that the cells are exposed to in the *in vitro* experiment. To improve the accuracy of the AED calculation, future studies using higher-tier models such as PBTK models that include different compartments or mass-balance models to calculate the mass distribution of a chemical within the *in vitro* test system at equilibrium, or that take into consideration metabolic activation, or experimental designs that take into account factors such as protein-lipid composition of the media, volatility properties of the chemical to reflect the true amount of free chemical in an *in vitro* assay (e.g. (gas-non permeable) plastic seals) could further optimize the IVIVE calculations (Armitage et al., 2014; Chang et al., 2022). As a consequence, the results generated using the generic HTTK model alone should be interpreted with caution.

Overall, whereas generic AEDs based on larger gene sets appear to be highly protective, biomarker AEDs might provide more predictive PoDs relative to hazard-associated *in vivo* PoD values for genotoxicants. However, some important considerations should be brought to bear. First, the number of chemicals included in our study was limited and more work with additional chemicals needs to be done to verify our conclusions. Second, although the uniformity and simplicity of the generic '3 compartments' model of the httk R package used in our study offer numerous advantages, this model has several limitations (Breen et al., 2021; Wetmore, 2015). For example, (i) the assumption of a constant dose rate of 1 mg/kg/day and a 100 % oral bioavailability may be considered overprotective or generally conservative from a human health standpoint to estimate the C_{ss}. Lower absorption or additional routes of excretion would result in a lower estimate of the AED required to achieve the C_{ss}. On the other hand, if active renal resorption occurs, it could result in a higher plasma C_{ss} for a given dose. With a higher-tier model, such as PBTK models that include different compartments or mass-balance models (e.g., equations that describe the rate of change of the amount of a substance in each compartment), discrete doses can be simulated to reach steady state (Pearce et al., 2017). (ii) Metabolites are currently not considered as this package only models metabolism as the elimination of the parent compound. Therefore, genotoxicants requiring metabolic activation should be assessed with more complex or

chemical-specific models. This might also explain the higher or equal AED values for the pro-genotoxicants NDMA and CPA in our study. (iii) Discrepancies may arise between the true amount of free chemical in an *in vitro* assay and the concentration used to derive AEDs, due to factors such as protein-lipid composition of the media, volatility properties of the chemical, and its affinity to bind to plastics (e.g., well-plate) (Blaauboer, 2010). Higher tier PBTK models may improve the accuracy of IVIVE and minimize the discrepancies between AEDs and PoDs (Beal et al., 2023). More specifically, despite its limitations, a generic TK model that is publicly available facilitates the application of IVIVE across a wide range of chemicals and enhances data reproducibility. Finally, another limitation of this study is the sole exposure time of 72 h, which was based on the experimental design of the GENOMARK biomarker. As highlighted by Carpi et al. (2024), PoDs derived from an *in vitro* concentration-response transcriptomics experiment may vary significantly with exposure duration, consistent with observations of toxicological studies involving whole organisms (Carpi et al., 2024). Hence, future *in vitro* transcriptomic studies may benefit from considering different exposure times to derive tPoDs and extrapolate to acceptable exposure levels in chronic exposure situations. Future studies may involve the application of our transcriptomic biomarkers to existing databases and the validation of these results with those of *in vivo* endpoints and/or long-term carcinogenicity studies.

5. Conclusion

In conclusion, we demonstrate a high level of concordance across *in vitro* PoDs and support their consistency in rank-ordering the potency of chemicals. We also reconfirm that the AEDs generally provide conservative PoDs relative to *in vivo* responses, with one exception (ENU). Our study extends previous research to increase confidence in the application of IVIVE to *in vitro* genotoxicity data, thereby supporting modern quantitative risk assessment practices without the use of animals. Prospective case studies can further help to investigate how to integrate transcriptomics in genotoxicity testing strategies.

CRediT authorship contribution statement

Mertens Birgit: Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Vanhaecke Tamara:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Rogiers Vera:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Van Goethem Freddy:** Conceptualization. **Beal Marc A.:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Yauk Carole L.:** Writing – review & editing, Conceptualization. **Meier Matthew J.:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Williams Andrew:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Cho Eunara:** Writing – review & editing, Data curation, Conceptualization. **Thienpont Anouck:** Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.tox.2025.154165](https://doi.org/10.1016/j.tox.2025.154165).

Data availability

All Templated Oligo-Sequencing (TempO-Seq) data are publicly available through NCBI Gene Expression Omnibus (GEO) under the accession number GSE281532.

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